

Lab Exercise: Patient Billing System

Objective:

To practice and apply Java programming concepts:

- Class and Object
 - Constructors
 - Methods
 - Passing parameters by object
 - `toString()` method
-

Task:

You are required to develop a simple **Patient Billing System** using two Java classes: `Patient.java` and `Hospital.java`.

`Patient.java` must include:

- Attributes:
 - `name (String)`
 - `id (String)`
 - `paymentAmount (double)`
 - A constructor to initialize all attributes.
 - Method `updatePayment(double newAmount)` to update the payment.
 - Method `copyPaymentFrom(Patient otherPatient)` to copy the payment from another patient (demonstrates passing an object).
 - Override `toString()` to return patient details in the format:
"Patient ID: P001, Name: Ali, Payment: RM 120.50"
-

`Hospital.java` must:

- Create two `Patient` objects with initial values.
 - Print both patients' details using `toString()`.
 - Update payment of the first patient.
 - Copy payment from the first patient to the second using `copyPaymentFrom()`.
 - Print updated patient details.
-

Sample Output:

Original Details:

Patient ID: P001, Name: Ali, Payment: RM 120.5

Patient ID: P002, Name: Siti, Payment: RM 80.0

After Changes:

Patient ID: P001, Name: Ali, Payment: RM 150.0

Patient ID: P002, Name: Siti, Payment: RM 150.0



Submission Guidelines:

Individual submission, but students are allowed to **discuss in the groups**.

Each student must:



Write **your own comments** in the code to explain each part.



Paste **Patient.java** and **Hospital.java** code into a **Microsoft Word** document.



Include a **screenshot of the output** (from IDE or command line).



Complete the **peer evaluation table** below.



Save the Word document as a **PDF** and submit.



Submission Checklist (PDF must include):



Patient.java code



Hospital.java code



Personal code comments (included in java programs)



Screenshot of output



Peer Evaluation Table

Peer Evaluation Table (5%)

Group Member Name	Idea (1.5)	Communication (1.5)	Programming Contribution (2)	Total (5)
WOO CHENG SHUAN	1.5	1.5	2	5
YASMIN BATRISYIA BINTI ZAHIRUDDIN	1.5	1.5	2	5

Patient.java

```
package PatientHospitalSystem;

public class Patient{
    private String name;
    private String id;
    private double paymentAmount;

    // Constructor to initialize attributes
    public Patient(String n, String i, double p) {
        this.name = n;
        this.id = i;
        this.paymentAmount = p;
    }

    public void setname(String name){
        this.name = name;
    }

    public void setid(String id){
        this.id = id;
    }

    public void setpaymentAmount(double paymentAmount){
        this.paymentAmount = paymentAmount;
    }

    public String getname(){
        return name;
    }

    public String getid(){
        return id;
    }

    public double getpaymentAmount(){
        return paymentAmount;
    }

    // Method to update payment amount
    public void updatePayment(double amount) {
        this.paymentAmount = amount;
    }
}
```

```
}

// Method to copy payment amount from another Patient object
public void copyPaymentFrom(Patient other) {
    this.paymentAmount = other.paymentAmount;
}

// Method to return a string representation of the Patient object
public String toString() {
    return "Patient ID: " + id + " , Name: " + name + " , Payment :
RM " + paymentAmount;
}

}
```

Hospital.java

```
package PatientHospitalSystem;

public class Hospital{
    public static void main(String[] args){
        // Create two Patient objects
        Patient patient1 = new Patient("Ali", "P001", 120.5);
        Patient patient2 = new Patient("Siti", "P002", 80.0);

        // Display original details
        System.out.println("Original Details: ");
        System.out.println(patient1.toString());
        System.out.println(patient2.toString());

        // Update payment for patient1 and copy payment from patient1
        // to patient2
        patient1.updatePayment(150.0);
        patient2.copyPaymentFrom(patient1);

        //Display updated details
        System.out.println("After changes: ");
        System.out.println(patient1.toString());
        System.out.println(patient2.toString());
    }
}
```

Screenshot of output

- PS C:\Users\User> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.14.7-hotspot\bin\java.exe' '-X X:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\User\AppData\Local\Temp\vscodesws_c6362\jdt_ws\jdt.ls-java-project\bin' 'PatientHospitalSystem.Hospital'
- Original Details:
Patient ID: P001 , Name: Ali , Payment : RM 120.5
Patient ID: P002 , Name: Siti , Payment : RM 80.0
After changes:
Patient ID: P001 , Name: Ali , Payment : RM 150.0
Patient ID: P002 , Name: Siti , Payment : RM 150.0